

Superradiant Thomson Scattering

Simulation Input Parameters & Physical Scales

Generated: 2026-09-17 14:27:24 UTC | Hartree a.u. ($c = 1/\alpha = 137.0360$)

1. Laser Pulse & Laguerre-Gaussian Mode Parameters

Parameter	Input Value	Unit	Resolved (a.u.)	Description
<code>laser.omega</code>	0.057	au	0.057	Laser angular frequency
<code>laser.a_0</code>	0.1	au	0.1	Normalized dimensionless vector potential
<code>laser.flat_top_periods</code>	10	au	10	Number of flat-top carrier periods
<code>laser.sigma_l</code>	2	T	220.463	Gaussian rise/fall duration parameter
<code>laser.wing_factor</code>	5	au	5	Gaussian wing truncation cutoff factor
<code>laser.p</code>	0	au	0	Laguerre radial mode index
<code>laser.m</code>	1	au	1	Laguerre azimuthal mode index (OAM charge)
<code>laser.epsilon</code>	-1	au	-1	Helicity / handedness parameter (+1 or -1)
<code>laser.w_0</code>	75	lambda	1.1329e+06	Beam waist radius at focal plane
<code>laser.zeta_x</code>	1	au	1	Polarization complex component x
<code>laser.zeta_y</code>	0 + 1j	au	0 + 1j	Polarization complex component y
<code>x_plot_laser</code>	0.5	w_0	5.6646e+05	Diagnostic point x coordinate
<code>y_plot_laser</code>	0	w_0	0	Diagnostic point y coordinate
<code>z_plot_laser</code>	0	w_0	0	Diagnostic point z coordinate

Derived Laser Physical Properties:

- Central wavelength: $\lambda = 1.5106 \times 10^4$ a.u. (799.36 nm)
- Optical period: $T = 1.1023 \times 10^2$ a.u. (2.67 fs)
- Total pulse duration: $\tau_{\text{pulse}} = 3.3069 \times 10^3$ a.u. (79.99 fs = 30.0 T)
- Beam waist radius: $w_0 = 1.1329 \times 10^6$ a.u. (59.95 μm)
- Rayleigh length: $z_R = 2.6694 \times 10^8$ a.u. (14125.81 μm = 17671.5 λ)
- Peak vector potential: $A_0 = 1.3704 \times 10^1$ a.u. ($a_0 = 0.1$)
- Peak electric field: $E_0 = 7.8111 \times 10^{-1}$ a.u.

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2. Electron Bunch & Initial Phase-Space Parameters

Parameter	Input Value	Unit	Resolved (a.u.)	Description
<code>electron.N</code>	512	au	512	Total number of electrons in ensemble
<code>electron.seed</code>	42	au	42	Random number generator seed
<code>electron.NT</code>	100	au	100	Trajectory sampling points per laser period
<code>electron.x_0</code>	0	w_0	0	Bunch center x coordinate
<code>electron.y_0</code>	0	w_0	0	Bunch center y coordinate
<code>electron.z_0</code>	0	w_0	0	Bunch center z coordinate
<code>electron.R_beam</code>	1.5	w_0	$1.6994e+06$	Cylindrical beam radius
<code>electron.h_beam</code>	0	w_0	0	Cylindrical beam length / height along Oz
<code>electron.px_beam</code>	0	c	0	Mean initial momentum px
<code>electron.py_beam</code>	0	c	0	Mean initial momentum py
<code>electron.pz_beam</code>	-1	c	-137.036	Mean initial momentum pz
<code>electron.sigma_px_beam</code>	0	c	0	Momentum spread standard deviation sigma_px
<code>electron.sigma_py_beam</code>	0	c	0	Momentum spread standard deviation sigma_py
<code>electron.sigma_pz_beam</code>	0	c	0	Momentum spread standard deviation sigma_pz

Derived Electron Bunch & Trajectory Properties:

- Mean initial Lorentz factor: $\gamma = 1.4142$
- Longitudinal velocity: $v_z / c = -0.7071$
- Relativistic Doppler factor: $D = (p^0 - p_z)/(mc) = 2.414214$
- Trajectory proper-time window: $\tau_{\max} = 1.3698e+03$ a.u. (33.13 fs)
- Proper-time ODE grid size: $N_{\tau} = 3001$ samples (100 points/period)
- Beam cylinder dimensions: $R = 1699386.66$ a.u., $h = 0.00$ a.u.

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3. Observation Screen Geometry & Evaluation Method

Parameter	Input Value	Unit	Resolved (a.u.)	Description
screen.shape	rectangular	au	rectangular	Detector geometry: rectangular or annular
screen.z_screen	-1.4400e+05	lambda	-2.1752e+09	Observation screen distance along Oz
screen.width	400	lambda	6.0423e+06	Rectangular screen width along Ox
screen.height	400	lambda	6.0423e+06	Rectangular screen height along Oy
screen.Nx	64	au	64	Screen pixel grid count along Ox
screen.Ny	64	au	64	Screen pixel grid count along Oy
screen.R_min	0	lambda	0	Annular screen inner radius
screen.R_max	400	lambda	6.0423e+06	Annular screen outer radius
screen.N_R	64	au	64	Annular screen radial ring count
screen.Phi_min	0	pi	0	Annular screen minimum angle
screen.Phi_max	2	pi	6.28319	Annular screen maximum angle
screen.N_Phi	64	au	64	Annular screen azimuthal sector count
screen.N_min	1	au	1	Minimum harmonic order evaluated
screen.N_max	3	au	3	Maximum harmonic order evaluated
screen.method	simplified	au	simplified	Field calculation method: 'simplified' or 'direct'

Observation Screen & Harmonic Frequencies:

- Screen distance: $z_screen = -2.1752e+09$ a.u. (-144000.0 lambda = -115.11 mm)
- Screen shape & pixels: rectangular (64 x 64 = 4096 pixels)
- Calculation formulation: simplified ('simplified' = Form 2, 'direct' = Form 1)
- Non-linear Thomson frequencies (Volkov dressed momentum):
 - Harmonic N=1: $\omega_1 = 0.330568$ a.u. (5.7994 $\omega_0 = 137.83$ nm)
 - Harmonic N=2: $\omega_2 = 0.661135$ a.u. (11.5989 $\omega_0 = 68.92$ nm)
 - Harmonic N=3: $\omega_3 = 0.991703$ a.u. (17.3983 $\omega_0 = 45.94$ nm)